

RESTful API Architecture, JWT Authentication & Modern Web

Subject: IT502 | Unit 3: Web Systems Architecture

1. Executive Overview & Syllabus Scope

Full-stack architectural guide covering REST constraints, HTTP methods and status codes, stateless JWT token authentication,

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Web Technologies & Distributed Web Systems
- 1. REST Constraints: Client-Server separation, Statelessness, Cacheability, Uniform Interface, Layered System
- 2. HTTP Semantics: GET, POST, PUT, PATCH, DELETE idempotency and response status code taxonomy
- 3. JSON Web Tokens JWT: Header, Payload, Signature construction and cryptographic verification
- 4. Web Security Essentials: Cross-Origin Resource Sharing CORS, CSRF tokens, XSS mitigation
- 5. API Scalability: Rate limiting, Reverse Proxies Nginx, Load Balancers, and Redis caching

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.